

Harvester v1.8.0 Installation SOP on KVM (Lab Environment)

1. Purpose

This document describes the procedure to deploy a single-node Harvester cluster on a KVM/libvirt environment for learning, testing, and proof-of-concept activities.

2. Environment Details

Component	Value
Hypervisor	KVM / Libvirt
Harvester Version	v1.8.0
Deployment Type	Single Node
CPU	24 vCPU
Memory	32 GB RAM
Disk	300 GB
Network	Bridge Network
Firmware	UEFI

3. Prerequisites

Hardware Requirements

Production

- CPU: 8+ Cores
- Memory: 64 GB RAM minimum
- Storage: 300 GB+

Lab

- CPU: 4+ vCPU
- Memory: 16–32 GB

- Storage: 150–300 GB
-

4. Download Harvester ISO

Download ISO from:

<https://github.com/harvester/harvester/releases>

Example:

harvester-v1.8.0-amd64.iso

Copy ISO to KVM host.

Example:

/var/lib/libvirt/images/harvester-v1.8.0-amd64.iso

5. Create VM Disk

Create 300 GB QCOW2 disk.

```
virsh vol-create-as Libvirt_Home harvesterdisk1 300G --format qcow2
```

Verify:

```
virsh vol-info harvesterdisk1 --pool Libvirt_Home
```

6. Create Virtual Machine

Recommended configuration:

Parameter	Value
vCPU	24
RAM	32 GB

Parameter	Value
Disk	300 GB
Firmware	UEFI
Network	Bridge

7. Attach ISO

Verify CDROM attachment:

```
virsh dumpxml harvester0 | grep source -A5
```

Expected:

```
<disk type='file' device='cdrom'>
<source file='/var/lib/libvirt/images/harvester-v1.8.0-amd64.iso' />
</disk>
```

8. Configure Boot Order

Edit VM XML:

```
virsh edit harvester0
```

Configure:

```
<os>
  <type arch='x86_64'>hvm</type>
  <boot dev='cdrom' />
  <boot dev='hd' />
</os>
```

9. Start Installation

Power on VM.

Select:

Harvester v1.8.0

Hardware check warning:

32GiB RAM detected
Harvester requires at least 64GiB RAM for production use

Select:

Yes

10. Installation Configuration

Provide:

Hostname

harvester.ocp4.imss.co.in

Management IP

Example:

172.16.3.57

DNS

8.8.8.8

Gateway

172.16.3.1

Disk

Select:

vda

Password

Set Admin Password

Example:

Infra123\$%^&

11. Installation Process

Harvester downloads and imports:

- Rancher Images
- RKE2 Images
- Harvester Components
- Longhorn Components

Installation may take:

20–40 minutes

depending on hardware.

12. First Boot

System reboots automatically.

Remove ISO after installation.

Either:

Cockpit

VM → CDROM → Eject

OR

Virsh

```
virsh change-media harvester0 sda --eject
```

13. Verify Node Status

Login to console.

Expected:

```
Harvester Cluster Status : Ready
Node Status : Ready
```

Example:

```
Management URL:
https://172.16.3.57

Node IP:
172.16.3.96
```

14. Access Harvester UI

Open browser:

```
https://172.16.3.57
```

Login:

```
Username: admin
Password: <Configured Password>
```

15. Verify Kubernetes Cluster

Login to node:

```
ssh rancher@172.16.3.96
```

Check cluster:

```
kubectl get nodes
```

Expected:

NAME	STATUS
harvester.ocp4.imss.co.in	Ready

Check pods:

```
kubectl get pods -A
```

16. Verify Rancher Components

```
kubectl get pods -A | grep rancher
```

Expected:

```
rancher
rancher-webhook
fleet-controller
fleet-agent
```

17. Verify API Server

```
kubectl cluster-info
```

Expected:

```
Kubernetes control plane is running at https://127.0.0.1:6443
```

18. Post Installation Tasks

Create VM Images

- Ubuntu
- RHEL
- Rocky Linux

Configure Storage

- Longhorn
- Additional Disks

Create Networks

- VLAN Networks
- VM Networks

Install Rancher

- Single Node Rancher
- Multi Cluster Management

Import Kubernetes Clusters

- RKE2
 - K3s
 - OpenShift
-

19. Troubleshooting

Boot Loop After Installation

Cause:

ISO still attached.

Solution:

```
virsh change-media harvester0 sda --eject
```

Cannot Login to UI

Reset password:

```
sudo passwd rancher
```

Or recreate admin password from Harvester node.

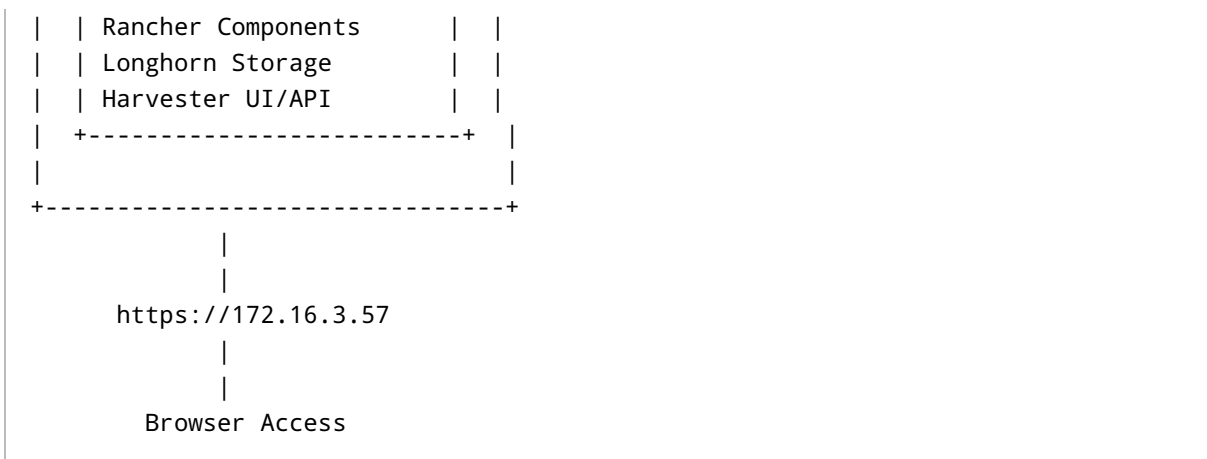
API Not Accessible

Check:

```
kubectl get nodes  
kubectl get pods -A
```

20. Architecture

```
+-----+  
| KVM Host |  
|         |  
| +-----+  
| | Harvester Node |  
| |-----|  
| | RKE2 Kubernetes |  
|         |
```



Installation Status

Deployment Type : Single Node Harvester

Version : v1.8.0

Cluster Status : Ready

Node Status : Ready

Management URL : <https://172.16.3.57>